St. Petersburg State University

Software Engineering Chair

Atomic Unit in Vortex GPU

Shamil Shaehmitov, SPBU SE

Saint Petersburg
2026

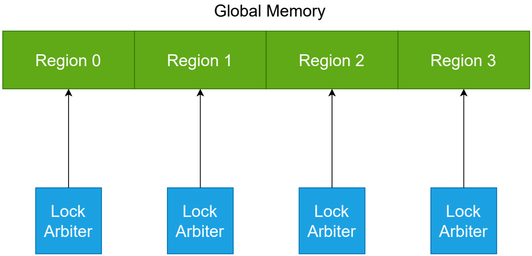
Introduction

- In the RTL design, support for extension RISC-V A was missing
- Support for these operations was added in PoCL
- A solution was developed that is currently under active development

Memory Lock Arbiters

- The memory was divided into regions, each of which was assigned an arbiter
- When one of the LSUs needs to perform an atomic operation in a region, the arbiter grants permission in a round-robin order
- The same arbiters are implemented for local memory if the core has more than one LSU
- The number of regions is configurable

Memory Lock Arbiters



Atomic Unit

- Placed in the LSU
- It is a finite state machine with 9 states
- States
  - IDLE: waiting for a request
  - LOCK: waiting for the lock to be granted
  - DRAIN: waiting for the pipeline to be drained
  - READREQ: sending a read request
  - READRSP: waiting for a read response
  - WRITEREQ: sending a write request
  - WRITERSP: waiting for a write response
  - FENCEREQ: sending a dummy read request before lock release
  - FENCERSP: waiting for a dummy read response

Optimization 1[WIP]

- Memory banks are processed sequentially.
- This makes it possible to complete all operations with fewer lock requests.
- It also requires entering the FENCE state less often.

Optimization 2[WIP]

- Skipping read and write operations for consecutive operations to the same address
- The read occurs during the first operation
- The write occurs during the last operation

LR/SC instructions

- LR/SC instructions are implemented using the same atomic unit
- The LR instruction is implemented as a read operation with a lock request
- The SC instruction is implemented as a write operation with a lock request
- Three register arrays of size SIMD_WIDTH are used to track the reservation: reservation validity, address, and data